

The effects of a 6-month exercise intervention programme on physical self-perceptions and social physique anxiety in non-physically active adolescent Swedish girls

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Abstract

Objectives: To examine the effects of a 6-month exercise intervention programme (EIP) on physical self-perceptions (PSP) and social physique anxiety (SPA) of non-physically active adolescent Swedish girls.

Methods: A true experimental design with randomization into an intervention or control group was used. The empowerment based EIP, offered to the intervention group twice a week for 6 months, consisted of 45-min exercise sessions followed by 15 min of discussions regarding a healthy lifestyle. A variety of exercise activities, chosen by the participants themselves, were used. Twenty-seven participants in the intervention group and 35 in the control group completed the Physical Self-Perception Profile (PSPP) and the Social Physique Anxiety Scale (SPAS) at pre- and posttest and, in addition, physical fitness, weight and height were measured.

Results and conclusions: The intent to treat analysis showed no significant improvements in PSPP subdomains, but lower SPAS scores for the intervention group, compared to the control group. However, when using a less conservative analysis, including only those who completed assessments both pre- and post-intervention, the intervention significantly reduced both PSPP subdomain and SPAS scores. The changes in PSPP and SPAS scores were not linked to changes in physiological variables. The results are put in the context of previous longitudinal and review studies, theoretical frameworks and models. The direction of effects, possible mechanisms and limitations of the study are discussed along with practical applications linked to exercise and modern diseases.

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Keywords: Exercise; Intervention; Physical self-perceptions; Social-physique anxiety

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The effects of a 6-Month Exercise Intervention Programme (EIP) on Physical Self-Perceptions and Social Physique Anxiety in Non-Physically Active Adolescent Swedish Girls

Physical self-perceptions (PSP) and self-esteem have been presented as core features of the pursuit of mental health and well-being (Fox, 1998; 2000a), both as mediators of other highly valued outcome variables such as happiness (Diener & Diener, 1995) and lack of mental illness (Baumeister, 1993), and as warranted outcomes (cf. Sonstroem, 1997; 1998; Sonstroem & Potts, 1996). In particular for girls/women, negative body image and low PSP have been linked to eating disorders, depression (see Cash & Pruzinsky, 2002; Levine & Piran, 2004; van de Vliet et al., 2002), and health-compromising behaviours such as smoking (Crocker et al., 2001). Hence, approaches to foster positive PSP and body image have been widely called for. In this context, exercise has been suggested as a significant and powerful instrument (Sonstroem, 1997; Sonstroem & Morgan, 1989).

Several cross-sectional studies have supported the association between PSP and self-reported levels of exercise (e.g., Fox and Corbin, 1989; Lindwall & Hassmén, 2004; Sonstroem, Harlow & Josephs, 1994). Unfortunately, as the majority of previous studies have employed ‘one-shot-cross-sectional designs’, important pieces of the puzzle of the complex dynamics of the physical self and the causal nature of effects between exercise and the physical self are still missing (e.g., Fox, 1998; 2000a). However, a longitudinal study investigating changes in PSP (perceived fitness) in adolescents over a 4-year period, demonstrated that, compared to a sedentary group, perceived fitness increased in physically active groups of girls over a 4-year period (Lintunen, 1995). Moreover, a review of exercise intervention studies, including 37 randomized controlled trials and 42 non-randomized trials, revealed that 78% of the studies indicated significant changes in self-perception and self-esteem outcome (Fox, 2000a). Well-designed intervention studies specifically focusing on the effects of exercise on PSP, however, are scarce despite the recommendations of several scholars (e.g., Fox, 1998; 2000a; 2000b; Sonstroem, 1997). Some studies have failed to find significant changes in PSP (Asçi, Kin & Kosar, 1998; Caruso & Gill, 1992), whereas others have found partial (Asçi, 2003) or clear support (Alfermann & Stoll, 2000; Page, Fox, McManus & Armstrong, 1993; Ransdell, Dratt, Kennedy, O’Neill, & DeVoe, 2001) for the effects of exercise on PSP. These previous studies also differed in design; some compared no exercise (control group) with exercise (e.g., Asçi, 2003) whereas others compared no exercise with different exercise programmes (e.g., Alfermann & Stoll, 2000), or different exercise programmes and/or lacked a control group (Caruso & Gill, 1992; Ransdell et al., 2001). The PSP-variables that have been identified as most susceptible to change through exercise are physical conditioning, physical strength and physical self-worth. Body attractiveness, in contrast, appears to be the subdomain least affected by exercise (see Fox, 1997; 1998). Supporting this notion, a recent prospective study found that changes in the physical condition subdomain predicted changes in physical activity in adolescent girls over a 12-month period (Crocker et al., 2003). In this context, the Exercise and Self-Esteem Model (Sonstroem & Morgan, 1989) and subsequently the expanded EXSEM (Sonstroem et al., 1994)—outlining how exercise and exercise interventions may affect PSP, thus leading to enhanced overall global self-esteem through increased physical self-efficacy—have been used as frameworks for describing the exercise-self-relationship (see Sonstroem, 1997; 1998).

Linked to the potential mechanisms of the exercise-PSP relationship, it has been suggested that changes in physiological fitness parameters are not necessary for enhanced self-perception through exercise, and that the mechanisms for change are likely to be psychosocial in nature (Fox, 2000b; Sonstroem, 1997; Taylor & Fox, 2005).

A widely investigated self-presentational concept that has also been related to exercise is social physique anxiety (SPA; Hart, Leary & Rejeski, 1989; Hausenblas, Brewer & Van Raalte, 2004; Leary, 1992).

This subtype of social anxiety denotes concern about others' negative evaluation of one's physique. SPA has been identified (Frederick & Morrison, 1996; Lantz, Hardy & Ainsworth, 1997) both as a potential barrier to exercise, due to concerns of revealing one's physique to others in an exercise setting and as an incentive, or motive, to take part in exercise as a means to decrease SPA through the development of a fitter and more attractive physique (see Hausenblas et al., 2004). Intervention studies over 5 months (McAuley, Bane & Mihalko, 1995), 10 weeks (Bartlewski, Van Raalte & Brewer, 1996) and 6 weeks (Williams & Cash, 2001) have demonstrated that exercise and circuit training may effectively reduce SPA. With regard to the link between PSP and SPA, Crocker et al. (2003) recently demonstrated that changes in PSP (body attractiveness) were associated with changes in SPA and dietary restraint in female adolescents.

Fox (2000a, 2000b) pointed out that the effects of exercise on self-concept are most likely to occur in those with initially low self-esteem or self-perceptions. In this context, women and girls have consistently demonstrated lower scores on PSP-instruments as well as elevated SPA in comparison to men and boys (e.g., Fox and Corbin, 1989; Hayes, Crocker & Kowalski, 1999; Hart et al., 1989; Lindwall & Hassmén, 2004) and may hence be targeted as a most relevant group for an exercise intervention.

Several limitations in previous research on the effects of exercise on psychological constructs, such as PSP and SPA, have been highlighted (e.g., Biddle, Fox, Boutcher, & Faulkner, 2000; Hausenblas et al., 2004; Leith, 1994; Morgan, 1997). The present study aimed to overcome some of the previous methodological shortcomings by: (a) adopting a true experimental design (i.e., a pre-post design) with randomization into control or experiment group using intent-to-treat analyses as a complement to standard analysis; (b) examining the effects over a longer period of time (i.e. 6 months); (c) using a relevant group of physically non-active adolescent girls, affording potential for change; and (d) measuring the dependent variables using valid and robust theory-driven instruments. Moreover, contrary to previous studies that often have used one type of exercise activity, the present study included a range of exercise activities, chosen by the participants themselves, conducted in a safe single-sex setting with the same exercise leader. The purpose of this strategy was to enhance feelings of control, competence and sense of belonging, all of which have been highlighted as potential mechanisms underpinning the exercise-physical self-relationship (e.g., Fox, 2000a), in addition to foster a positive motivational climate through feelings of enjoyment and self-motivation.

Hence, the purpose of the present study was to investigate the effects of a 6-month EIP on PSP and SPA for physically non-active adolescent girls. Based on previous research vis-à-vis the design of the current study, it was hypothesized that: (1) the intervention group would demonstrate enhanced PSPP- and SPAS- scores after 6 months compared to the control group; (2) the largest effects on PSP would be shown for the PSPP-variables physical conditioning, physical strength and physical self-worth, whereas body attractiveness would demonstrate the smallest effects (3) the changes in PSPP- and SPAS-scores would not be linked to changes in physiological fitness parameters.

Method

Participants

The participants consisted of non-physically active adolescent girls recruited from seven schools that are representative of low socio-economic areas in two communities in the south of Sweden. As low socio-economic status consistently has manifested itself as a significant determinant of physical

inactivity for adolescents as well as for adults (see [Buckworth, 2000](#)), these areas were chosen in order to identify non-physically active adolescent girls. Schools were matched according to enrollment size, urban or suburban location and socio-economic status of area to ensure a homogeneous sample. The school-nurses and physical education teachers from the schools, having the knowledge of the physical activity status of the non-physically active girls, contacted them and offered them the chance to participate in the study. In total, 180 non-physically active girls accepted the invitation to participate in the study and attended an information meeting. One hundred ten of these were eligible according to the inclusion criteria. Hence, these 110 girls provided baseline measures and were then randomized. The age of the participants ranged from 13 to 20 (M: 16.35; SD: 1.56). Mean weight, height, and BMI at pretest were 62.75 kg (SD=12.34), 1.65 m (SD=0.07), and 23.05 kg/m² (SD=4.06), respectively. The inclusion criteria for participating in the study were: (1) less than 20 min of self-reported physical activity per day (e.g. walking to and from school); (2) exercise less than once a week in leisure time; (3) non-physically active for at least one year; and (4) medically healthy in terms of (a) not being insulin dependent, (b) not suffering from hypertension, or (c) not taking any form of beta-blocker. The theoretical basis for the inclusion criteria was based on Swedish national guidelines and international research recommendations ([Folkhälsoinstitutet, 1999](#); [US Department of Health and Human Services, 1996](#)). The majority of the participants (70%) reported previous participation in a sports club. Similarly, 45% reported a history of regular exercise.

Instruments

Physical self-perception profile

Physical self-perceptions were measured by means of the PSPP-inventory ([Fox, 1997](#); [Fox and Corbin, 1989](#)). The PSPP is a 30-item self-report questionnaire consisting of five sub-domains, each of which has six items. The sub-domains are: Perceived (1) Sport Competence (Sport), (2) Physical Conditioning (Condition), (3) Bodily Attractiveness (Body), (4) Physical Strength (Strength), and (5) Physical Self-Worth (PSW). Two alternative statements or descriptions of people are presented, from which the individuals can choose the one which best describes themselves, ranging from 'sort of true' to 'really true'. Each item is then scored from 1 to 4. In previous research, the PSPP has proved sensitive to a wide range of individual differences but has not appeared to be influenced by social desirability ([Fox, 1998](#); [Fox and Corbin, 1989](#)). Internal consistency scores have typically ranged from 0.83 to 0.92 for men and women, although the subscale Sport has sometimes yielded a lower alpha (for example 0.70 as reported by [Hayes et al., 1999](#)). In the present study, one item in the Strength subdomain was deleted due to low item reliability in baseline data, resulting in a 5-item Strength subscale. Exploratory and confirmatory factor analyses have supported the factor validity of the PSPP across various samples, age groups (see [Fox, 1998](#)) and cultures ([Hagger, Lindwall & Asci, 2004](#)).

Translation of the PSPP into Swedish was conducted according to a standardised back-translation procedure. The original PSPP was first translated into Swedish and sent to two bilingual translators (Swedish–English), who translated it back into English. Differences regarding the versions were discussed and resolved with the sport psychology research group at the university. The final version was tested in a pilot study on university students, which resulted in acceptable psychometric attributes with Cronbach's alphas values from 0.86 to 0.93 ([Lindwall & Johnson, 2001](#)). Support for the factor validity of the Swedish version of PSPP has recently been demonstrated ([Hagger et al., 2004](#)). As the Swedish version of the PSPP had not been used on the particular population of this study (i.e., young adolescents)

pretesting procedures (c.f. Bergman, 1995), including interviews and discussions with a group of non-physically active adolescents' girls, was conducted by the second author before the start of the intervention to evaluate the participants understanding of the items. This procedure led to a minor revision of the instrument to reduce the risk of low content validity in the specific population of the study. Internal consistency values for the subscales in the pre-test and post-test were: 0.76–0.74 (Sport); 0.62–0.65 (Condition); 0.84–0.84 (Body); 0.74–0.77 (Strength); 0.86–0.85 (PSW).

Social physique anxiety scale

The Social Physique Anxiety Scale (Hart et al., 1989) was employed to measure SPA. The respondents are asked to rate 12 statements on a 5-point Likert type scale. The anchors are 'not at all' (1), 'slightly' (2), 'moderately' (3), 'very' (4), and 'extremely' (5). Examples of items are: 'I am comfortable with the appearance of my physique' and 'Unattractive features of my physique/figure make me nervous in certain social settings'. Previous research with SPAS has demonstrated test-retest reliability and internal consistency with Cronbach's alpha values of around 0.90 (Hart et al., 1989).

The Swedish translation of the SPAS was conducted according to a back-translation process method where two bilingual (English-Swedish) translators translated the instrument into Swedish and then back into English. Differences were discussed and resolved by the sport and exercise-psychology group at a university. The Swedish version of the SPAS has demonstrated acceptable alpha values ranging from 0.89 to 0.94 (Lindwall, 2004a; Lindwall & Johnson, 2001). The present study adopted the empirically supported 7-item unidimensional scale (Lindwall, 2004a). However, as item 4 ('there are times when I am bothered by thoughts that other people are evaluating my weight or muscular development negatively') was divided into two items for the purpose of clarity (one focusing on weight and the other on muscular development), this resulted in an 8-item scale. The same pre-testing procedures that was conducted for the PSPP was also used for the SPAS before the start of the study to ensure a high content validity for the specific population. Internal consistency ranged from 0.85 at pre-test to 0.89 at post-test.

Body composition and physiological fitness

Measurements of weight and height were collected in order to calculate Body Mass Index (BMI). The Åstrand submaximal work test with the bicycle ergometer was used to predict submaximal oxygen uptake as a measure of physical fitness according to standard procedures (cf. Åstrand, 1976).

Procedure

The study adopted a true experimental design with the schools, and hence participants in groups, being randomized into an intervention ($n=56$) or control group ($n=54$). The randomization of schools—resulting in that peers from the same school were allocated to the same group—had two purposes: (a) to avoid contamination between participants in the exercise group and control group, and (b) to increase motivation and adherence in the exercise group through the creation of a safe social environment involving familiar people. The randomisation process was conducted blindly by the second author (who was not involved in the intervention) before the baseline assessments, using sequentially numbered, and sealed, envelopes. The participants in the intervention group took part in an exercise intervention programme (EIP), which ran from early Autumn 2002 to mid Spring 2003. The participants in the control group were put in a waiting-list, which meant that they were offered the opportunity to take part in the EIP after 6 months. No forms of alternative activities were organized for the control group.

The study was conducted in accordance with the ethical guidelines of the Swedish Humanistic-Social Science Research Board. Written and oral information that participation was voluntary, that data would be treated confidentially and that participants were free to withdraw at any time, was provided. All variables were assessed before and after the EIP (i.e. at pre- and post-test). Questionnaires and physical fitness tests were completed after school hours. Questionnaires were distributed and the physical tests were conducted by research assistants in a test room with groups of two to four participants. The research assistants gave oral instructions to the participants on how to fill in the questionnaire and were also available to answer questions while the questionnaire was being filled in.

The EIP, run by the Halland District Sport Federation, was designed to increase moderate physical activity, PSP, and body image in non-physically active adolescent girls through a variety of activities, influenced by the participants themselves. The intervention was based on the empowerment process (Gibson, 1991; cf. Lindgren, 2002) and was designed to promote the participants abilities to meet their own needs and to mobilize the necessary resources in order to strengthen their beliefs in their ability to participate in different sports without feeling ashamed of their body or their capability. Hence, the EIP activities were structured to foster a sense of self-control, pride concerning the own body, and feelings of competence. These factors are believed to be important mechanism's underpinning the effect of exercise on PSP. The EIP was structured in a single-sex setting to avoid the potentially inhibitory effects of young males on the participant females with regards to their PSP and body image and to create a context where the participants could bond with one another, fostering sense of belonging. The empowerment process in the EIP also emphasized enjoyment of exercise, in contrast to the focus on performance, seriousness and a competitive setting of typical sport-club settings.

Several strategies to maintain a high adherence in the EIP over the intervention period were used. For instance, the participants were offered a chance to select numerous sport activities they enjoyed to meet their specific needs and interests and to increase their sense of involvement. Also, if the participants did not attend to the EIP for four consecutive sessions (i.e., two weeks), they were contacted on phone and asked why they had not been participating and reminded that they were welcomed back.

The EIP consisted of four groups, each with its own exercise leader. The groups were homogeneous in terms of age and residential location. In order to create similar supportive atmospheres in the different groups, the intervention staff had three meetings with the exercise leaders, two before the start of the EIP and one in the middle of the EIP, discussing the EIP in order to ensure that the leaders did not manage the EIP in different ways.

The intervention program was offered twice a week for a 6-month period (except for two weeks during Christmas holiday) and included: (a) exercise (45 min) and; (b) discussion (15 min) where issues such as healthy lifestyles (including healthy dietary and physical activity behaviours) were addressed. The activities chosen by the participants in the intervention groups included aerobics, water aerobics, step-up, badminton, kick-boxing, spinning, dancing, climbing, bowling, karate, jujitsu, yoga and different ball-games which favored a sense of belongingness, group-cohesion and successful learning experiences. The intervention staff who lead the exercise and discussion sessions, were exercise instructors and assistant coaches in the sports clubs.

Statistical analyses

A MANOVA, univariate ANOVAs, and chi-square tests were conducted to check for baseline equivalence between the groups. Violations of assumptions for the ANOVAs was checked. A two-way

(Group X Time) mixed univariate ANOVAs with Group (intervention vs. control) as between subject factor were used to analyze the effects of the exercise intervention on SPAS scores. A repeated measure multivariate ANOVA with Group as between subject factor was used to examine the impact of the intervention on the five PSPP scales (dependent variables). A GLM analysis of covariance, adjusting for pretest scores, was used to investigate post-test differences between the experiment and the control group. Zero-order correlations were used to examine the relation between changes in physiological variables and changes in PSPP- and SPAS-scores. In addition to these analyses, intent-to-treat analysis (see [Hollis & Campbell, 1999](#)) was also performed to examine the effects of the EIP on the PSPP- and SPAS-variables. The rationale for conducting the intent-to treat-analysis was that previous reported studies have typically only included those participants that followed through the intervention, thereby risking inflating any overall effect (see [Biddle et al., 2000](#); [Taylor & Fox, 2005](#)). The use of intent-to treat analysis, which includes all participants that were originally randomized in the study design, is a more conservative approach and less likely to overestimate any potential effects. The method used to handle missing data involved imputation of values from the previous assessment with complete data (i.e., values from pre-test were carried forward and used as post-test values).

Results

Sixty-two of the 110 (57%) participants completed both the pre- and post-test questionnaires. Twenty-seven of these were in the intervention group and 35 in the control group. The average number of sessions attended for the participants in the intervention group for the exercise activities was 56% (SD:19%). Univariate ANOVAs showed that there were no differences in any of the PSPP or SPAS-variables at pre-test ($p > 0.05$) between drop-outs and participants who completed the intervention. Descriptive data of the intervention and control groups are presented in [Table 1](#). Univariate ANOVAs showed that the groups did not differ in terms of weight, height, BMI or physical fitness at the pretest ($p > 0.05$). Moreover, Chi square tests demonstrated that the participants in the intervention and control groups did not differ with regard to parents' education level, type of residence and living area (living in an area of apartment houses or in a residential area), parents' exercise habits, previous membership of and participation in sports clubs, or previous exercise participation ($p > 0.05$). Univariate ANOVA confirmed that the SPAS-scores of the participants in the two groups did not differ at pre-test ($p > 0.05$). A MANOVA revealed that the groups did not differ with regards to the five PSPP scales (Pillai's Trace = 0.08, $F(5,60) = 0.75$, $p > 0.05$).

Table 1
Baseline characteristics (M and SD) of all participants and study completers

Variables	All participants		Study completers	
	Intervention ($n = 56$)	Control ($n = 54$)	Intervention ($n = 27$)	Control ($n = 35$)
Age (years)	15.3 (1.9)	15.5 (1.1)	15.2 (1.7)	15.4 (1.0)
Weight (kg)	63.0 (14.6)	62.5 (9.5)	63.2 (16.9)	62.2 (10.0)
Height (m)	1.64 (0.07)	1.66 (0.07)	1.63 (0.06)	1.66 (0.06)
BMI	23.5 (4.9)	22.6 (2.9)	23.5 (5.2)	22.9 (3.3)
Physical fitness ^a	36.8 (8.8)	37.0 (6.6)	38.1 (10.3)	37.6 (7.3)

^a Submaximal oxygen uptake test score.

Table 2

Pre- and post-test scores and effect sizes (η^2) of post-test differences between the intervention and the control group for the SPAS and PSPP with intent-to-treat analysis

Variables	Pre-test		Post-test		η^2
	M	SD	M	SD	
SPAS					
Intervention ^a	24.4	6.3	23.9	7.4	0.06
Control ^b	24.0	6.1	25.4	6.9	
PSPP					
Sport					
Intervention	13.5	3.7	14.3	3.5	0.02
Control	13.2	3.9	13.3	3.9	
Condition					
Intervention	12.3	2.8	13.7	3.3	0.08
Control	13.4	3.4	14.0	3.3	
Body					
Intervention	13.6	4.5	14.2	4.9	0.00
Control	15.2	5.1	15.5	5.4	
Strength					
Intervention	10.1	3.3	10.9	3.4	0.01
Control	9.6	3.5	10.1	3.6	
Physical self-worth					
Intervention	13.3	4.3	14.1	4.7	0.00
Control	14.5	4.2	14.6	4.1	

Significant differences between groups at post-test controlling for pre-test scores; $p < 0.05$.

^a $n = 56$.

^b $n = 54$.

Using an intent-to-treat analysis, the MANOVA for the PSPP variables showed a significant Time effect (Pillai's Trace=0.18, $F(5,104)=4.63$, $\eta^2=0.18$, power=0.97, $p < 0.01$) and non-significant effects for Group (Pillai's Trace=0.09, $F(5,104)=2.03$, $\eta^2=0.09$, power=0.66, $p > 0.05$) and Group X Time interaction (Pillai's Trace=0.04, $F(5,104)=0.80$, $\eta^2=0.04$, power=0.28, $p > 0.05$). No univariate ANOVAs for the PSPP variables were significant. The Time X Group interaction effect was significant for SPAS (Pillai's Trace=0.06, $F(1,108)=6.33$, $\eta^2=0.06$, power=0.70, $p < 0.05$). Adjusting for pre-test scores, using a GLM analysis of covariance, significant differences (see Table 2) on adjusted post-test means were found only for the SPAS, ($F(1,107)=6.19$; $p < 0.05$).

Using less conservative analyses, (i.e., only including participants that followed through the programme in the analyses), the main effect for Time for SPA was not significant (Pillai's Trace=0.02, $F(1,60)=0.97$, $\eta^2=0.02$, power=0.16, $p > 0.05$). However, a significant Time X Group interaction effect was found (Pillai's Trace=0.11, $F(1,60)=7.34$, $\eta^2=0.11$, power=0.76, $p < 0.01$). The intervention group showed lower levels of SPA at post-test, whereas the control group reported elevated SPAS-scores (see Table 3). For the PSPP subscales, MANOVA demonstrated a significant effect for Time (Pillai's Trace=0.37, $F(5,56)=6.50$, $\eta^2=0.37$, power=1.00, $p < 0.01$) and a non-significant effect for Group (Pillai's Trace=0.10, $F(5,56)=1.29$, $\eta^2=0.10$, power=0.42, $p > 0.05$) but no significant interaction effect. Univariate ANOVAs revealed significant Time effects for Sport ($F(1,60)=6.70$; $p < 0.05$), Condition ($F(1,60)=29.49$; $p < 0.01$), Strength ($F(1,60)=13.40$; $p < 0.01$),

Table 3

Pre- and post-test scores and effect sizes (η^2) of post-test differences between the intervention and the control group for the SPAS and PSPP

Variables	Pre-test		Post-test		η^2
	M	SD	M	SD	
SPAS					
Intervention ^a	26.3	8.2	24.0	8.8	0.08
Control ^b	24.4	6.6	25.6	7.7	
PSPP					
Sport					
Intervention	12.9	3.5	14.5	3.3	0.06
Control	13.5	3.7	13.7	3.5	
Condition					
Intervention	12.2	2.6	14.9	3.2	0.08
Control	13.4	3.5	14.4	3.4	
Body					
Intervention	13.7	4.6	14.9	5.3	0.01
Control	15.1	4.8	15.4	5.3	
Strength					
Intervention	9.4	3.0	11.0	3.4	0.03
Control	9.4	3.2	10.1	3.5	
Physical self-worth					
Intervention	14.4	4.2	16.2	4.2	0.09
Control	13.9	3.9	14.0	3.8	

Significant differences between groups at post-test controlling for pre-test scores; $p=0.05$; $p<0.05$.

^a $n=27$.

^b $n=35$.

and PSW ($F(1,60)=5.21$; $p<0.05$) in addition to a non-significant trend for Body ($F(1,60)=3.24$; $p=0.08$). A non-significant trend for the interaction main effect of Time X Group on the PSPP subscales was detected (Pillai's Trace = 0.15, $F(5,56)=1.95$, $\eta^2=0.15$, power = 0.61, $p=0.10$). Follow-up ANOVAs showed significant interaction effects for Sport ($F(1,60)=3.90$; $p<0.05$), Condition ($F(1,60)=7.45$; $p<0.01$), and PSW ($F(1,60)=4.36$; $p<0.05$), indicating that the intervention group increased more than the control group on these subscales.

Adjusting for pre-test scores, significant differences on adjusted post-test means (see Table 3) were found on: SPAS ($F(1,59)=6.41$; $p<0.05$); Sport ($F(1,59)=3.65$; $p=0.05$); Condition ($F(1,59)=5.16$; $p<0.05$); and PSW ($F(1,59)=5.62$; $p<0.05$). Compared to the control group, participants in the intervention group revealed lower scores on SPAS and higher scores on the three PSPP-subscales at post-test.

To investigate whether the changes in PSPP and SPAS scores were related to changes in physiological parameters, correlations between physiological change scores (i.e. post- and pre-test measurements) of weight, BMI and physical fitness, and changes in PSP and social physique anxiety scores were examined for both groups combined. The only significant correlations were found between Condition change and weight ($r=-0.30$, $p<0.05$), BMI ($r=-0.29$, $p<0.05$), and physical fitness ($r=0.27$, $p<0.05$). When separately investigating these correlations for each group, no significant correlations were found.

Mean adherence rate in for the exercise activities in the intervention group was not significantly correlated to the change scores of any of the PSPP or the SPAS variables.

Discussion

As stated by Crocker et al. (2003): ‘Change analysis moves us one step closer to establishing causal claims between variables and within settings that are difficult to manipulate experimentally’ (p. 335). Accordingly, this study examined the effects of a 6-month EIP on PSP and SPA for adolescent girls, using a true experimental design with repeated measurements. In accordance with the first hypothesis, the results overall support the positive effect of exercise on PSP and SPA, with the intervention group showing a positive, or more positive, change in PSP and SPA compared to the control group where the changes were negative (for SPA), or less positive (for PSP). However, as a caveat, when using intent-to-treat analysis, the effects of the intervention did not remain significant for any of the PSPP-variables. Hence, from a more strict and conservative analytical perspective, the support for the positive effects of the exercise intervention on the physical self in the present study is not so clear. This signifies the need for researchers in future intervention-studies to conduct an intent-to-treat analysis before examining any effects for only those that stay in the treatment, in order to limit potential threats to internal validity.

However, using a less conservative analysis, the PSP results agree with longitudinal studies (Lintunen, 1995), the meta-analyses of Fox (2000a), and provide further support for the EXSEM-model (e.g., Sonstroem, 1997; Sonstroem et al., 1994). Moreover, supporting the second hypothesis of the present study, the largest effect sizes in the present study were accounted for by the Condition- Sport- and PSW subdomains, whereas Body showed the smallest change. As the main focus of the EIP programme and the activities therein, were on general perceptions of the physical-worth, sport competence in addition to physical conditioning, the detected changes on these particular subdomains seems quite natural. Also, these results agree strongly with previous work (e.g., Fox and Corbin, 1989; Lindwall & Hassmén, 2004), and highlight Condition and Body as the two PSPP subdomains most respectively least related to exercise. Explaining these consistent results, it has been suggested (e.g., Fox, 1997; Lindwall, 2004b; Lindwall & Hassmén, 2004) that the definition and operationalization of Condition are more closely linked to the physical aspect of the exercise than to the other PSPP subdomains. The Body sub-domain addresses appearance components as opposed to perceived competence and is probably more difficult to change by means of exercise because people find it less easy to control and the outcome evaluation being based on more diffuse criteria than the other sub-domains.

Moreover, these results corroborate some previous intervention studies (e.g., Alfermann & Stoll, 2000; Bartlewski et al., 1996; McAuley et al., 1995; Page et al., 1993; Ransdell et al., 2001; Williams & Cash, 2001), and contradict others (e.g., Asçi, 2003; Asçi et al., 1998; Caruso & Gill, 1992). As these previous intervention studies are very heterogeneous in terms of design and participant characteristics it is important to compare the results of the present study foremost with the previous studies that have focused solely on young adult females and compared exercisers with no exercisers (i.e., Asçi, 2003; Asçi et al., 1998; Bartlewski et al., 1996). In this context, the results of the present study clearly support the positive effects of exercise on SPA for young females reported by Bartlewski et al., (1996). However, they clearly contradict the Asçi et al. (1998) study and, to some extent, also the Asçi (2003) study; both on the effects of exercise on PSP. Two factors that may explain the different and somewhat contradictory results on the effects of exercise on PSP in this study and previous ones are: (a) that this study examined

changes over a longer period of time (6 months vs. 10 and 8 weeks), which has been targeted as a vital component for demonstrating reliable changes for psychological constructs resulting from exercise (e.g., Leith, 1994); and (b) that the present study focused on the effects of exercise participation with a plethora of activities chosen by the participants themselves, using interlinked discussions for feed-back, and a more natural as well as non-evaluative settings compared to previous studies, which employed structured exercise sessions focusing on a single activity per study (e.g., aerobic dance- or strength-training). Also, the EIP-program of this study was specifically structured to target important mechanisms underpinning the effects of exercise on PSP, including sense of control, feelings of competence and belonging (Fox, 2000a; 2000b).

Linked to the overall effect of the intervention, it should also be mentioned that, in general, the effect sizes in the study were small, with no η value larger than 0.09. Hence, although the less conservative analysis showed that significant decreases in PSPP and SPAS scores did take place, the magnitude of these changes are still small and their impact on other important behavioural, emotional and cognitive aspects of every-day-life for the girls are uncertain.

The reduction in SPA-scores (being significant also when using intent-to-treat analysis) for the intervention group compared to the control group, indicates that exercise, in combination with continuous critical discussions of the role and function of the body, may foster positive feelings of competence, self-worth and acceptance in regards to the body, especially in girls/women (cf. Cash, 1997; Fredrickson & Roberts, 1997). In this context, the empowerment-based design of the exercise intervention, aimed at fostering self-control, decision-making and activity (e.g., letting the girls themselves influence the choice of activities) and conducted in a single-sex setting, presumably contributed to the important development of a non-evaluative exercise environment, minimizing self-presentational concerns and focusing on positive body image and PSP (see Hausenblas et al., 2004). In contrast to other studies showing that participants in control groups reduced their SPAS-scores to the same degree as participants in various exercise classes, (Diehl & Petrie, 1995), the control group in the present study revealed elevated SPAS-scores after 6 months, further highlighting the relevance of the documented beneficial effects of exercise on SPA for the intervention group. Moreover, given the acknowledged role of PSP and body image as significant mediators in the sport-self-esteem relationship (Richman & Shaffer, 2000) - indicating that sport participation for females will only promote self-esteem if it also fosters vital mediating variables such as PSP and body image - the results of this study may suggest a potential for enhancement of global self-esteem for the girls in the intervention group.

With regard to the direction of effects, the interaction between exercise and the self may be viewed from two perspectives: the skill development hypothesis and the self-enhancement hypothesis (Marsh, 1986; Sonstroem, 1997; 1998). Due to its design, the present study focused on, and its results support, the skill development hypothesis suggesting that exercise participation results in increased self-perceptions and self-esteem due to feelings of competence and well-being associated with exercise- as opposed to the self-enhancement hypothesis, which holds that individuals tend to seek out activities that are in accord with their self-perceptions in order to demonstrate their competence, i.e. high PSP and self-esteem lead to exercise participation. However, these two hypotheses should be viewed as complementary rather than exclusive, and most probably work in parallel or simultaneously to influence the exercise behaviour of the individual. For example, a person starting an exercise program and experiencing the accompanying positive feelings of competence and improved PSP or self-esteem, may, subsequently, continue to seek out, or maintain the exercise activity in order to retain his/her level of PSP.

Linked to the direction of effects, it is also relevant to draw attention to the underlying mechanisms associated with the effects of exercise on PSP and SPA. Suggested explanatory factors are: an undetermined psychophysiological factor, improvements in physiological fitness or weight loss, increased sense of effectiveness, autonomy, personal control, and self-acceptance, and an enhanced sense of belonging and significance (Fox, 2000a; 2000b). Similarly, various potential mediators of the positive effects of exercise on SPA have been proposed (e.g., Bartlewski et al., 1996; Hausenblas et al., 2004): familiarity with the exercise setting and repeated exposure to sources of anxiety, social support, physiological changes such as weight loss and increased muscle tone, and positive changes in other closely related constructs (e.g. the self-concept) being associated with SPA-changes. In the present study, changes in PSP and SPA were only weakly related to changes in the physiological parameters of body weight, BMI and physical fitness. However, considering that this relation was only found for one variable (Condition) when both groups were examined together, the results of the study in general support the third hypothesis of the study and previous work (Fox, 2000a; Knapen et al., 2003; Sonstroem, 1997), suggesting that the mechanisms of change in the physical self are psycho-social, rather than psychophysiological, in nature. However, several of the aforementioned psycho-social mechanisms probably interacted to create the positive effects on the physical self observed in this study. For instance, group dynamic factors such as belongingness and group cohesion, which were vital components in the EIP, may have contributed to the demonstrated effects in self-perceptions for the intervention group. In addition, different mechanisms probably affect different individuals in a complex pattern, making a clear-cut interpretation difficult. With regard to potential mechanisms, the role of empowerment (see Lindgren, 2002) in the positive relationship between exercise and body image/physical self should also be discussed. Generally, as an important ingredient in the intervention design and aiming at fostering feelings of self-control and self-activation, feelings of empowerment may have created beneficial interaction effects with the exercise experiences on the physique-related cognition and perceptions of the girls. Hence, exercise appear highly relevant in the empowerment process, which involves enhancing people's abilities to meet their own needs and solve their own problems, in particular adolescent girls. However, with reference to the quotation from Crocker et al. (2003) at the beginning of this discussion, it should be noted that the settings surrounding the physical self and exercise are difficult to manipulate experimentally (e.g., Ojanen, 1994). Hence, the possibility that factors other than the exercise intervention may have influenced the results should also be considered. Two such factors, associated with placebo effects and expectancies of the experimenter, are the Rosenthal effect and demand characteristics (cf. Morgan, 1997). However, as it is next to impossible to discern the true effects from placebo effects in exercise intervention studies (Ojanen, 1994), no clear-cut conclusion can be drawn in terms of effects attributable to other factors. However, given that the differences in effects in the various subdomains are in accordance with previous work, it seems reasonable to conclude that possible placebo effects only account for a minor part of the demonstrated effects of exercise on the physical self in this study (assuming that nuisance factors do not systematically co-vary with the different PSP-subdomains).

The drop-out rate of over 40% should be acknowledged as a limitation of the study. Although drop-out analyses revealed no pre-test differences between drop-outs and participants remaining at post-test, an insufficient number of participants in each group may result in power problems in the statistical analyses and thereby a risk of type II-error (e.g., Cook & Campbell, 1979). The different results in the present study found when using less conservative analytical procedures compared to when using intent-to-treat analysis further emphasize the need to incorporate all participants that were originally

randomized in the analysis instead of only, as typically been done, examining the effects of the most highly motivated participants that follow through the intervention. However, as this drop-out rate is fairly common in intervention studies (and also closely corresponds to the widely reported 50% rate of natural drop-out in exercise programs over a 6 month period; e.g., Dishman, 1988), researchers should also incorporate the concepts of practical and clinical significance (see Ogles, Lunnen & Bonesteel, 2001) into their analyses in future studies and link the changes in scores to relevant behavioural changes (see Stoové & Andersen, 2003) rather than focusing solely on statistical significance, in order to evaluate the meaningfulness of change (see Morgan, 1997).

Moreover, as the present study, and the design therein, targeted a specific population, i.e. previously physically non-active adolescent females from low socio-economic areas, the results of the study should foremost be generalized to this population.

From a research perspective, the results of the present study show that broader exercise programmes, lasting over a longer period of time, involving different activities that the participants choose themselves, structured on psychological factors rather than physiological parameters such as intensity, may have the same, or larger, effects as shorter, more structured single-activity interventions. Also, the present study highlights the value of structuring interventions in research in ways that make them similar to the context, settings and processes of every day-life exercise. This issue is important, not least regarding the external validity of the research conducted, when communicating with the outside world and ‘spreading the word’ to, for instance, clinical psychology and health services (see Fox, 2000b).

Future studies should investigate the suggested mechanisms underpinning the exercise-physical self-relationship from an interdisciplinary approach (see Lindwall, 2004b) and evaluate the effects of different types of exercise interventions, based on different mechanisms, on different populations. Also, if possible (see Ojanen, 1994) researchers should design studies that may separate the effects of the intervention and independent variable from various extraneous factors, such as various forms of placebo-effects. For instance, well-controlled RCT studies with exercise placebo-groups over a longer period of time, with follow-up assessments, would give us increased knowledge of the exercise-self-relationship.

From an applied perspective, bearing in mind the negative health implications (i.e. depression, eating disorders and smoking) associated with low self-esteem and low PSP, especially for girls, in mind, the results of this and previous studies are highly relevant. Given the growing evidence of the beneficial effects of exercise on the self and the physical self (e.g., Fox, 2000a), practitioners should be aware of, and if appropriate implement, exercise as a complement to ordinary treatment, such as medication and therapy (see Levine & Piran, 2004) when finding strategies to combat the increasing mental health problems (e.g., Biddle et al., 2000) and the general disrespect for the body, especially among young girls and women.

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